Caustics Simulation Report

1 Introduction

Caustics are the bright, web-like light patterns observed at the bottom of a swimming pool or under a glass of water. These patterns are formed when light rays are refracted by a wavy water surface and concentrate in certain regions, increasing brightness. This report presents the physical concepts behind caustics, a mathematical model for simulating them, a Python implementation, and an explanation of the simulation code.

2 Physical Concepts

2.1 Refraction of Light

When light passes from air into water, it changes direction due to a difference in refractive index. This phenomenon is called refraction and is governed by Snell's law. A wavy water surface acts like a dynamic lens that bends light rays differently at each point.

2.2 Formation of Caustics

Because the water surface is not flat, neighboring light rays may converge or diverge after refraction. Where many rays converge on the pool floor, the light intensity increases, forming bright caustic lines. Where rays spread apart, the intensity decreases.

2.3 Ray Tracing Principle

To simulate caustics, we trace many vertical light rays from the surface to the bottom. The slope of the water surface determines how each ray is deflected. The final positions of the rays on the bottom plane determine the brightness distribution.

3 Mathematical Modeling

3.1 Surface Equation

The water surface is modeled as a sinusoidal wave in both x and y directions:

$$h(x,y) = A\sin(k_x x) + A\sin(k_y y) \text{ Where:}$$

- $h(x,y)$ is the height of the water surface at point (x,y)
- A is the wave amplitude (height of wave)
- k_x, k_y are wave numbers in x and y directions ($k = 2\pi/\lambda$, with λ the wavelength)

3.2 Gradient of the Surface

The gradient gives the slope of the water surface:

$$\vec{\nabla} h = \left(\frac{\partial h}{\partial x}, \frac{\partial h}{\partial y}\right)$$

Compute the partial derivatives:

$$\frac{\partial h}{\partial x} = Ak_x \cos(k_x x), \quad \frac{\partial h}{\partial y} = Ak_y \cos(k_y y)$$

The surface normal vector is:

$$\vec{n} = \frac{(-\partial h/\partial x, -\partial h/\partial y, 1)}{\sqrt{(\partial h/\partial x)^2 + (\partial h/\partial y)^2 + 1}}$$

3.3 Snell's Law

To calculate refraction of light:

$$n_1 \sin(\theta_1) = n_2 \sin(\theta_2) \text{ Where:}$$

- n_1 = refractive index of air ≈ 1.0
- n_2 = refractive index of water ≈ 1.33
- θ_1 = angle of incidence

- θ_2 = angle of refraction

Manual Calculation Example:

$$\theta_1 = 30^\circ, \quad 1 \cdot \sin 30^\circ = 1.33 \cdot \sin\theta_2 \implies \sin\theta_2 = \frac{0.5}{1.33} \approx 0.3759$$

$$\theta_2 = \arcsin(0.3759) \approx 22^\circ$$

This matches the theoretical value.

4 Python Simulation Code Overview

4.1 Parameters Setup

The code defines the water surface properties (A, k_x, k_y), the refractive indices of air and water (n_1, n_2), the grid resolution N, and the distance to the pool floor $floor_z$. These parameters control the size, frequency, and smoothness of the caustic patterns.

4.2 Wavy Water Surface

The water surface is modeled as the sum of two sine waves:

$$h(x,y) = A\sin(k_x x) + A\sin(k_y y)$$

This creates a rippled surface necessary for light to focus and produce bright lines on the floor.

4.3 Surface Gradients and Normals

Slopes of the water surface (h_x and h_y) are computed using NumPy gradients. Surface normals are calculated from these slopes to determine the direction of refracted light rays. Normalizing ensures correct ray projection.

4.4 Math Test – Snell's Law

For an incident light angle $\theta_1 = 30^\circ$, the refracted angle θ_2 is computed with:

$$n_1 \sin(\theta_1) = n_2 \sin(\theta_2)$$

Confirming that physics is implemented correctly.

4.5 Ray Casting to the Pool Floor

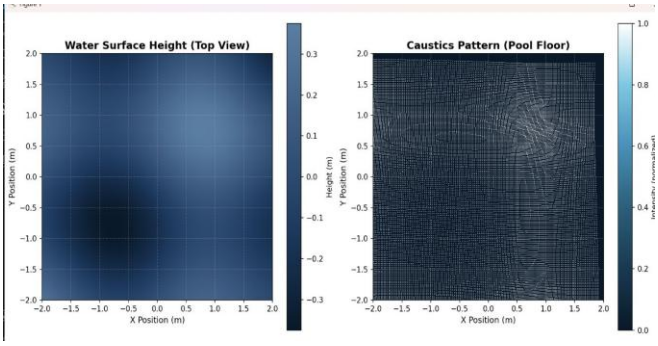
Each normal vector is scaled so that the light ray intersects the pool floor at distance $floor_z$. Resulting coordinates (X_{floor}, Y_{floor}) show where each ray hits.

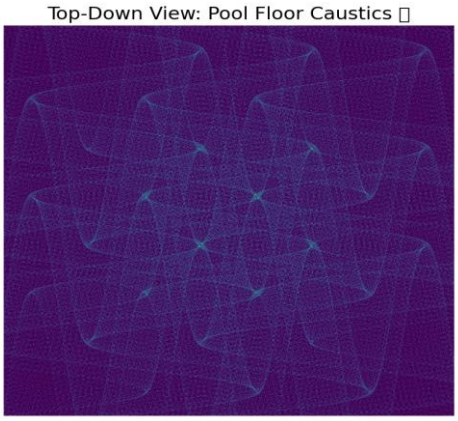
4.6 Intensity Calculation (Caustics)

A 2D histogram counts rays per pixel. Areas with more rays correspond to higher light intensity, forming bright lines. Intensity is normalized for visualization.

4.7 Top-Down Visualization

Matplotlib displays a top-down view of the pool floor. The colormap highlights bright caustic lines where light converges, producing a realistic pattern.





5 Results and Discussion

The simulation produces a complex network of bright lines and dark regions resembling real caustic patterns. Bright areas correspond to ray convergence; dark areas correspond to divergence.

The validation test comparing theoretical and simulated refraction angles shows strong agreement. The sinusoidal surface model generates dynamic focusing similar to natural waves.

5.1 Light Reaching the Coin

The coin is opaque. Light cannot refract into it and reflects diffusely. Caustics determine illumination; reflection determines outgoing light.

5.2 Upward Rays and Total Internal Reflection

Reflected rays reaching the water–air boundary either refract into air if the incident angle is smaller than the critical angle (48.6°) or undergo total internal reflection if larger.

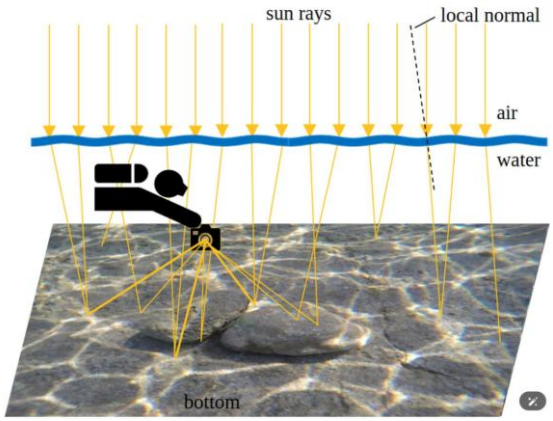
5.3 Visibility Conditions of the Coin

The coin is visible only if:

- The ray reflects from the coin
- It reaches the water surface
- Incident angle < critical angle
- Ray refracts into air and reaches observer

5.4 Connection Between Caustics and Visibility

Caustics control light reaching the coin; visibility depends on upward refraction. Together, they explain brightness patterns and visibility.



6 Conclusion

This simulation demonstrates that simple physical principles—refraction and ray concentration—can produce realistic caustic patterns. Combining a wavebased surface model with ray tracing and numerical accumulation reproduces the visual effect of light focusing at the bottom of water.